

represent the second dimension. The *max()* and *min()* functions are used to return the maximum and minimum values respectively.

```
> a <- array(c(1, 2, 3, 4, 5, 6, 7, 8, 9), dim=c(3, 3))
> a
      [,1] [,2] [,3]
[1,]    1    4    7
[2,]    2    5    8
[3,]    3    6    9

> apply(X=a, MARGIN=1, FUN=max)
[1] 7 8 9
> apply(X=a, MARGIN=2, FUN=max)
[1] 3 6 9
> apply(X=a, MARGIN=1, FUN=min)
[1] 1 2 3
> apply(X=a, MARGIN=2, FUN=min)
[1] 1 4 7
```

If you would like to return a list, you can use the *lapply()* function to a vector or list, as shown below:

```
> l = list(1, 3, 5, 7)
> lapply(l, function(x) x*x)
[[1]]
[1] 1
[[2]]
[1] 9
[[3]]
[1] 25
[[4]]
[1] 49
```

On the other hand, if you would like to return a vector, matrix or array, you can use the *sapply()* function.

```
> sapply(l, FUN=function(x) x*x)
[1] 1 9 25 49
```

The *mapply()* function is a multivariate version of the *sapply()* function. For example:

```
> mapply(paste, c("1", "2", "3", "4"),
+        c(".", ".", ".", "."),
+        c("Chapter: Introduction", "Chapter: Trees",
+         "Chapter: Graphs", "Chapter: Networks"))
      1 2
"1 . Chapter: Introduction" "2 . Chapter: Trees"
      3 4
"3 . Chapter: Graphs"     "4 . Chapter: Networks"
>
```

The following arguments are supported by the *mapply()* function.

- *FUN*: The function to be applied to the data.
- *...*: A list of vectors on which the function should be applied.
- *MoreArgs*: Additional arguments to the function.
- *SIMPLIFY*: A Boolean value on whether to simplify the result.
- *USE.NAMES*: A Boolean value on whether to use names for the values.

Select

The *shingle()* function can group data into bins. You will need to load the *lattice* library in order to use the function. For example, the various cars in 'new_mtcars' are grouped based on 4, 6, or 8 cylinders, as indicated below:

```
> library(lattice)
> shingle(new_mtcars$cy1)
```

```
Data:
[1] 6 6 4 6 8 6
```

```
Intervals:
  min max count
1   4   4     1
2   6   6     4
3   8   8     1
```

```
Overlap between adjacent intervals:
[1] 0 0
```

A list of discrete factors is returned from a continuous numerical vector using the *cut()* function. The horsepower in *new_mtcars* data is broken into two groups, as follows:

```
> cut(new_mtcars$hp, breaks=2)
[1] (92.9,134] (92.9,134] (92.9,134] (92.9,134] (134,175]
(92.9,134]
Levels: (92.9,134] (134,175]
```

The *cut()* function accepts the following arguments:

- *x*: A numeric vector.
- *breaks*: The number of points to split the data into.
- *labels*: The labels for the factor levels.
- *include.lowest*: A Boolean value on whether to include the smallest value in the bin.
- *right*: A Boolean value on whether the interval should be open on the left and closed on the right.

The bracket notation can be used on the data frame to filter the results. For example, the cars that have more than five cylinders are listed below:

```
> new_mtcars[new_mtcars$cy1>5,]
```